

A multi-view subspace representation learning approach powered by subspace transformation relationship

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ABSTRACT

Traditional multi-view subspace representation learning typically captures the pairwise similarity among data objects but neglects the underlying relationship among subspace representations of different views, resulting in the essential structural information of multi-view data not being fully exploited. In this study, a multi-view representation learning approach is proposed using subspace transformation relationships. First, a subspace transformation relationship is defined to extract the underlying subspace relationship information. The transformation relationship is constrained by a low-rank term to enhance the consistency of the relationships and is used to construct a multi-view subspace representation model. Second, an optimization method is proposed for the multi-view subspace representation model using the ALM-ADM strategy. Third, a novel multi-view subspace representation learning algorithm is presented using the model and optimization method. Consequently, experimental findings on benchmark datasets indicated that the algorithm outperformed the state-of-the-art algorithm, benefitting from extracting subspace transformation relationships to guarantee high-order correlation information from multi-view data.

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1. Introduction

Multi-view data are collected from various sources or represented by different feature descriptors. Each type of feature is considered as a view to describe the data objects from different perspectives [1–5]. Multi-view representation learning is a compelling topic in machine learning that aims to obtain comprehensive representations of different views [6–11]. A comprehensive representation is crucial for downstream tasks such as clustering and classification [12–15]. The primary challenge in multi-view representation learning is to ensure consistency among various views when utilizing the local structural information of individual views.

Existing multi-view subspace representation learning methods project multi-view data into a common underlying low-dimensional subspace to explore complex correlations in multi-view data [14,16–18]. The learned subspace representation matrices primarily reflect the relationships between data objects, whether consistent or complementary [19–23]. Furthermore, the relationship at the sample level expresses only the data correlation in a single view. However, the subspace representations of

different views are derived from the same data items. Different representations of the same data must be transformed into each other. For example, the visual and voice representations of TV shows can be transformed into each other. An underlying transformation relationship is inferred to be implied between the subspace representations that is a higher-order relationship than that between data objects. However, the underlying transformation relationship among the subspace representations, which benefits high-order correlations, is neglected. Consequently, the essential structural information of multi-view data has not been fully exploited. Therefore, the underlying subspace transformation relationships in the multi-view subspace representation learning process must be explored. In this study, a novel multi-view representation learning algorithm (MSRLSTR) is proposed with a subspace transformation relationship term to extract the relationships among subspaces and a low-rank constraint term on the relationships for consistency. A manifold constraint is also used to acquire the intrinsic construction of a single view for complementary purposes.

The main contributions of this study are as follows:

- A subspace transformation relationship was defined to extract the underlying subspace transformation relationship information.
- A multi-view subspace representation model was constructed using the subspace transformation relationship constraint.

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Table 1
Symbols and their description.

Symbol	Notation	Symbol	Notation
n	Number of samples	T^{ij}	Relationship between views i and j
V	Number of views	$A^v \in R^{d_v \times n}$	Lagrange multipliers
$X^v \in R^{d_v \times n}$	Data matrix of the v th view	$\ \cdot\ _*$	Nuclear norm
$Z^v \in R^{n \times n}$	Self-representation matrix	$\ \cdot\ _F$	Frobenius norm

- An alternating optimization method for the subspace representation model was proposed using the ALM-ADM strategy.
- A novel multi-view representation algorithm was proposed using the subspace representation model.

2. Related works

Multi-view representation learning aims to generate an appropriate representation that reflects the fundamental structure of multi-view data, mainly using alignment-based and fusion-based methods [5,14].

Alignment-based methods, including canonical correlation analysis (CCA) and its extensions, fuse multiple feature sets into a unified latent space [12,13,24,25]. CCA aims to merge different views for consistency [26,27]. Using neural networks to investigate nonlinear correlations, DCCA expands CCA [12,28]. Accordingly, the alignment-based methods maintain consistent features. However, the complementary information of a single view was neglected, degrading the learned representation.

Fusion-based methods integrate representations from multiple views into a single view using subspace representation learning and neural networks [29–33].

Subspace representation learning methods. Subspace representation learning methods fuse multiple feature sets into a common space to preserve consistent and diverse relationships. The LT-MSL introduces a low-rank tensor constraint to analyze the high-order correlation information from different views of data objects [34]. DiMSC enforces diverse representations among different views [19]. Certain methods based on DiMSC have been proposed to explore consistent and complementary information [20,35]. PMSC and COMVSC are partition-level multi-view learning approaches that can learn a view representation individually and defuse the influence of other views [22,36]. The TMSRL is a semi-supervised algorithm that obtains a high-level relationship regarding the subspace representation matrices as low-rank tensors [37]. CMRL uses the subspace representations and low-rank constraints to explore representations with consistency and complementarity [23]. However, the exploration of subspace transformation relationships has been neglected in the current methods.

Neural network-based methods. *AE²Nets* integrated information using two autoencoders [38]. CVE-AE is a cross-view transformation equivariance auto-encoder for correlation preservation across different views [15]. KGAN boosts the multi-view representation model using a knowledge graph for sentiment analysis [39].

In summary, the unique information of a particular view is generally ignored in alignment-based methods. Subspace-based approaches focus on maintaining correlations of samples. However, subspace representation methods neglect the correlation relationship between views.

3. Multi-view subspace representation learning

Certain preliminaries are briefly introduced in this section with a general formulation of subspace representation learning. Symbols and the corresponding descriptions listed in Table 1 are used in this study for better understanding.

3.1. Subspace representation learning

Subspace representation learning extracts the underlying subspace projections of the original data. The derived self-representation matrix comprises pairwise similarities between data objects. Each sample is expressed as a linear aggregation of the remaining samples [20,40,41]. A self-representation matrix is often obtained by determining the following objective function:

$$\min_{Z^v} = \sum_v \|X^v - X^v Z^v\|_F^2 + \Omega(Z^v), \quad (1)$$

where $\sum_v \|X^v - X^v Z^v\|_F^2$ represents the loss in subspace representation learning. $\Omega(Z^v)$ represents the regularization term for self-representation Z^v that varies with the approach.

3.2. Manifold regularization

To obtain a self-representation matrix with the intrinsic local information of an individual view, a manifold constraint is used to retain the smoothness of the representation [19,35]. The manifold constraint is formulated as

$$\Omega(Z^{(v)}) = \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n w_{ij}^{(v)} \|z_i^{(v)} - z_j^{(v)}\|_2^2 = \text{tr}(Z^{(v)} L^{(v)} Z^{(v)T}) \quad (2)$$

tr denotes the trace. $W^{(v)} = (w_{ij}^{(v)})$ can be attained using an inner product that measures the proximity of objects. $L^{(v)}$ represents a Laplacian matrix representing the difference between $D^{(v)}$ and $W^{(v)}$. $D^{(v)}$ represents the diagonal degree matrix. $d_{ii}^{(v)} = \sum_{j=1}^n w_{ij}^{(v)}$. The purpose of Eq. (2) is to reduce the disparity between the representations of the data objects that are close to each other in the original space. Therefore, the manifold term can maintain the local smooth information of the data objects [19].

4. Proposed subspace representation model

To obtain a representation with the correlation subspace transformation relationship information and local information of a single view, a novel multi-view subspace representation model was constructed (shown in Fig. 1) using manifold, transformation relationship, and low-rank constraints.

As shown in Fig. 1, the transformation relationship constraint (T^{ij}) is used to describe the subspace transformation relationship information. A rank constraint is used on the relationship matrix ($\|T^{ij}\|_*$) to regulate the consistency of the relationships. In addition, the manifold constraint, as described in Eq. (2), is exploited for the intrinsic information of a single view. Consequently, both consistent and intrinsic information is acquired.

4.1. Subspace transformation relationship

The subspace representation matrices learned in the existing subspace representation learning methods focus on the pairwise relationships among data objects [19,21,22]. Subspace representations of different views are extracted from the same data objects and indicated the underlying transformation relationships. In other words, different representations of the same data

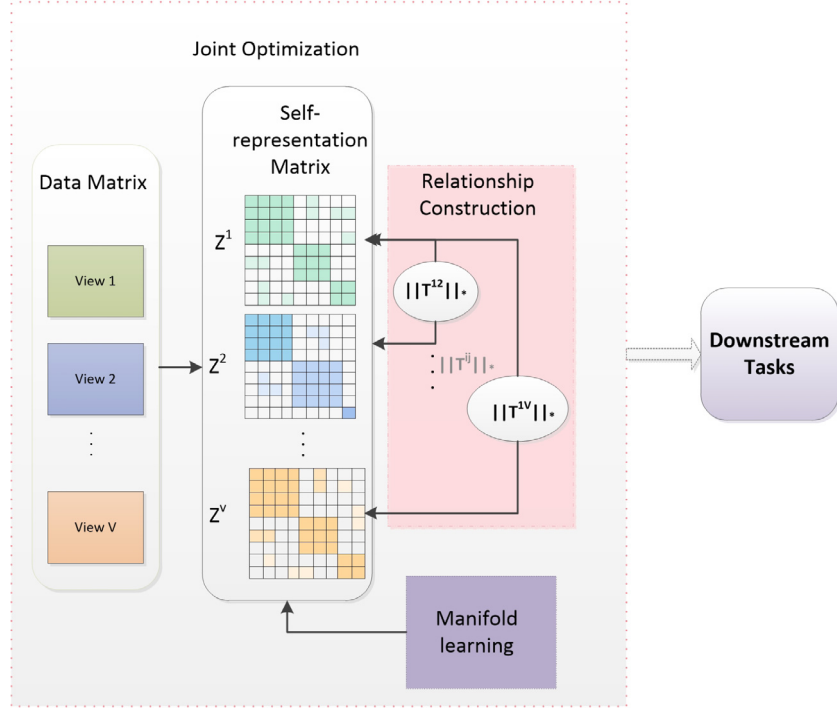


Fig. 1. The framework of MSRLSTR.

must be transformed into one another. However, the underlying transformation relationships have been neglected. The underlying relationship is a higher-order relationship than those between data objects that is beneficial in guaranteeing the essential structural information. To explore the transformation relationship information of the subspace representations in subspace representation learning, a subspace transformation relationship constraint is defined as follows:

Definition 1 (Subspace Transformation Relationship T). For any subspace representation matrix U and V ($U, V \in R^{n \times n}$), the goal of the subspace transformation relationship is to determine a matrix T ($T \in R^{n \times n}$), such that, $TU \approx V$.

Definition 1 shows that subspace transformation relationship T is a bridge between any two subspaces U and V and expresses the relationship between subspaces U and V .

In Eq. (1), $\Omega(Z^v)$ represents the regularization term for controlling the fitting of the self-representation Z^v . To guarantee the underlying relationships among the subspace representation matrices of different views (Z^v), the following subspace transformation relationship can be employed on the subspace representation learning model in Eq. (1).

$$\min_{Z^v, T} = \sum_v \sum_{k \neq v} \|T^{vk} Z^v - Z^k\|_F^2 \quad (3)$$

s.t. $v, k = 1, 2, 3, \dots, V (v \neq k)$,

where Z^v denotes the subspace representation of the v th view. T^{vk} denotes the subspace transformation relationship between Z^v and Z^k .

Subspace transformation representation T in Eq. (3) represents the relationship among the subspace representations for exploring higher-order information that is deficient in the representation learned through the pairwise relationships among

data objects. Therefore, the subspace transformation relationship term retains high-level consistency and promotes subspace representation.

4.2. Low-rank constraint term

To reinforce the correlation of transformation relationships T^{vk} , a low-rank constraint is employed on T^{vk} . This is because a relationship with a low rank generally has high consistency [40, 42]. The low-rank constraint on T^{vk} is formalized as follows:

$$\min_T = \sum_v \sum_{k \neq v} \|T^{vk}\|_* \quad (4)$$

s.t. $v, k = 1, 2, 3, \dots, V (v \neq k)$.

Because of the nonconvexity of the rank constraint, a nuclear norm served as a convex approximation employed on T^{vk} to ensure the high consistency of representation matrix Z^v .

4.3. Objective formulation

Accordingly, to produce the subspace representation with intra-view information (Eq. (2)) and inter-view subspace transformation relationships constrained by low-rank (Eqs. (3), (4)), the objective function is induced as follows:

$$\begin{aligned} \min_{Z^v, T} = & \sum_v \|X^v - X^v Z^v\|_F^2 + \lambda_1 \sum_v \text{tr}(Z^v L^v Z^{vT}) \\ & + \lambda_2 \sum_v \sum_{k \neq v} \|T^{vk} Z^v - Z^k\|_F^2 + \lambda_3 \sum_v \sum_{k \neq v} \|T^{vk}\|_* \quad (5) \end{aligned}$$

s.t. $v, k = 1, 2, 3, \dots, V (v \neq k)$,

where $\lambda_1 > 0$, $\lambda_2 > 0$, and $\lambda_3 > 0$ are trade-off factors that are used to balance the inference of manifold, subspace transformation, and rank constraints.

In Eq. (5), manifold constraint $\text{tr}(Z^v L^v Z^{vT})$ is used to maintain the local information of a single view. Subspace transformation relationship T^{vk} is used to retain the high-level relationship of the self-representation matrix. The rank constraint on the relationship matrix T^{vk} is used to maintain consistent information between multiple views. Therefore, the correlation (consistency) and independence (complementarity) of the multi-view data can be simultaneously obtained (as presented in Fig. 1).

5. Multi-view subspace representation model optimization

5.1. ALM-ADM strategy and model optimization

The augmented Lagrangian multiplier with alternating direction minimization (ALM-ADM) can optimize Eq. (5) that has coupling variables. ALM-ADM is widely used in linearly constrained optimization problems, particularly to decouple multivariable coupling optimization problems. First, we disentangle Eq. (5) into some parts. Then, we address the subproblems in parallel. Finally, we coordinate the results of the subproblems to obtain the overall result [43].

To decompose Eq. (5), we propose auxiliary variables $Y = T$. The objective function can be expressed as follows:

$$\begin{aligned} \min_{Z^v, T} &= \sum_v \|X^v - X^v Z^v\|_F^2 + \lambda_1 \sum_v \text{tr}(Z^v L^v Z^{vT}) \\ &+ \lambda_2 \sum_v \sum_{k \neq v} \|T^{vk} Z^v - Z^k\|_F^2 + \lambda_3 \sum_v \sum_{k \neq v} \|Y^{vk}\|_* \\ \text{s.t. } &T^{vk} = Y^{vk}, \quad v, k = 1, 2, 3, \dots, V (v \neq k). \end{aligned} \quad (6)$$

The ALM function can be expressed as follows:

$$\begin{aligned} L(Z^v, T^{vk}, Y^{vk}) &= \sum_v \|X^v - X^v Z^v\|_F^2 + \lambda_1 \sum_v \text{tr}(Z^v L^v Z^{vT}) \\ &+ \lambda_2 \sum_v \sum_{k \neq v} \|T^{vk} Z^v - Z^k\|_F^2 \\ &+ \lambda_3 \sum_v \sum_{k \neq v} \|Y^{vk}\|_* \\ &+ \sum_v \sum_{k \neq v} \Psi(A^{vk}, T^{vk} - Y^{vk}), \end{aligned} \quad (7)$$

$\Psi(G, J) = \mu \|J\|_F + \langle G, J \rangle$, where $\mu > 0$ is a penalty scalar, and $\langle \cdot, \cdot \rangle$ reveals the matrix inner product. $A^{vk} (k \neq v)$ are the Lagrange multipliers.

In the following, we obtain the optimization of Z^v , T^{vk} , and Y^{vk} .

Z^v -subproblem: To optimize variable Z^v in Eq. (7), we fix variables T^{vk} and Y^{vk} . Eq. (8) is the optimization objective function for Z^v .

$$\begin{aligned} \min_{Z^v} &= \sum_v \|X^v - X^v Z^v\|_F^2 + \lambda_1 \sum_v \text{tr}(Z^v L^v Z^{vT}) \\ &+ \lambda_2 \sum_v \sum_{k \neq v} \|T^{vk} Z^v - Z^k\|_F^2. \end{aligned} \quad (8)$$

To obtain Z^v , we obtain the derivative of Eq. (8) with regard to Z^v equal to 0. The result of the optimization of Z^v is as follows:

$$\left(X^{vT} X^v + \lambda_2 \sum_{k \neq v} T^{vkT} T^{vk} \right) Z^v + \lambda_1 Z^v L^v = X^{vT} X^v + \lambda_2 \sum_{k \neq v} T^{vkT} Z^k \quad (9)$$

Eq. (9) is a typical Sylvester equation that has a unique solution [44].

Y^{vk} -subproblem: To optimize variable Y^{vk} in Eq. (7), we fix variables Z^v and T^{vk} . Eq. (10) is the optimization objective function for Y^{vk} .

$$\begin{aligned} \min_{Y^{vk}} &= \lambda_3 \sum_v \sum_{k \neq v} \|Y^{vk}\|_* + \sum_v \sum_{k \neq v} \Phi(A^{vk}, T^{vk} - Y^{vk}) \\ &= \min_{Y^{vk}} \frac{\lambda_3}{\mu} \sum_v \sum_{k \neq v} \|Y^{vk}\|_* + \sum_v \sum_{k \neq v} \|Y^{vk} - \left(T^{vk} + \frac{A^{vk}}{\mu}\right)\|_F \end{aligned} \quad (10)$$

We can obtain Eq. (11) according to Si et al. [20] and Luo et al. [40].

$$Y^{vk} = S_{\frac{\lambda_3}{\mu}} \left(T^{vk} + \frac{A^{vk}}{\mu} \right) \quad (11)$$

where $S_{\tau}[\cdot]$ represents the singular value shrinkage operator [45].

T^{vk} -subproblem: To optimize variable T^{vk} in Eq. (7), we fix variables Z^v and Y^{vk} . Eq. (12) is the optimization objective function for T^{vk} .

$$\min_{T^{vk}} = \lambda_2 \sum_v \sum_{k \neq v} \|T^{vk} Z^v - Z^k\|_F^2 + \sum_v \sum_{k \neq v} \Phi(A^{vk}, T^{vk} - Y^{vk}) \quad (12)$$

To obtain T^{vk} , we obtain the derivative of Eq. (12) for $T^{vk} = 0$. The result of the optimization of T^{vk} is as follows:

$$T^{vk} = \left(2\lambda_2 Z^k Z^{vT} + \mu Y^{vk} \right) \cdot \left(2\lambda_2 Z^v Z^{vT} + \mu I \right)^{-1} \quad (13)$$

Multipliers A^{vk} and μ : Multipliers A^{vk} and μ are updated using the following formulas:

$$A^{vk} = A^{vk} + \mu (T^{vk} - Y^{vk}) \quad (14)$$

$$\mu = \min(\mu_{\max}, \mu \rho) \quad (15)$$

5.2. Multi-view subspace representation algorithm

According to the variable optimization described in Section 5.1, the proposed idea of multi-view subspace representation learning with a subspace transformation relationship and manifold constraint can be obtained, as shown in Algorithm 1.

Algorithm 1: MSRLSTR

Input : Initialize coefficient matrices $Z = T = Y = 0$, $\mu = 10^{-6}$, $\mu_{\max} = 10^6$, $\rho = 1.1$. Maximum iterations $Iter = 100$ and stop threshold $\varepsilon = 10^{-6}$.

Output: representation matrix Z^v

```

1 for t ∈ [Iter] do
2   for v ∈ views do
3     Update Zv by Eq.(9);
4     for k ∈ views do
5       Update Yvk by Eq.(11);
6       Update Tvk by Eq.(13);
7       Update the multipliers Avk and μ by Eqs.(14), (15);
8       if converges then
9         break;
10      end
11    end
12  end
13 end
```

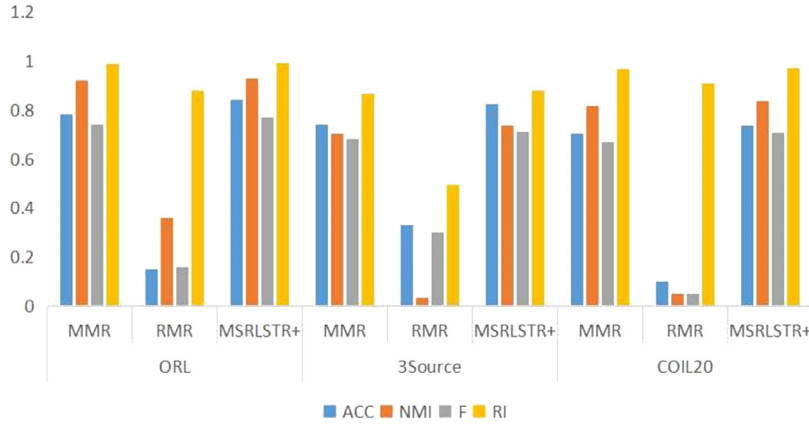


Fig. 2. Ablation study of MSRLSTR+ on three datasets.

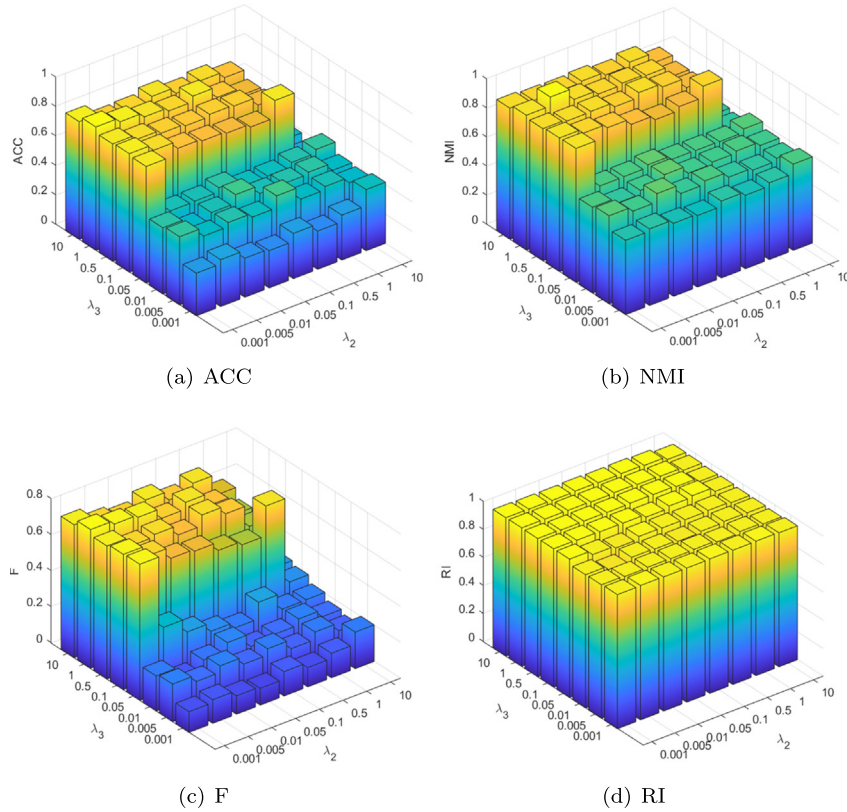


Fig. 3. Effect of different values of λ_2 and λ_3 for clustering performance on the ORL.

The proposed algorithm consists of three subproblems, as shown in Algorithm 1. The complexity of the subproblem Z^v is $O(n^3)$ according to Cao et al. [19] and Tan et al. [35], where n denotes the number of samples. Because of the matrix inversion, the cost of updating subproblem T^{vw} is $O(n^3)$. Revising subproblem Y^{vw} is $O(n^3)$ in complexity owing to the nuclear norm. Therefore, the overall complexity is $O(tv^2n^3)$, where t and v denote the numbers of iterations and views, respectively.

6. Experiments

To evaluate the performance of the proposed method, we conducted a comparative trial, parameter sensitivity test, and ablation experiment. Windows 11 was the operating system used in the trial, along with an Intel Core i7-10510U processor clocked

at 1.80 GHz and 8 GB of RAM. We implemented the proposed algorithm on the Matlab2018a platform.

We conducted experiments using five popular benchmark datasets. **ORL** contained 400 samples that are consistent with the facial images of 40 people. Intensity, LBP, and Gabor were used to extract the characteristics of the views. **COIL-20** included 20 categories. A total of 1440 pictures were extracted using intensity, LBP, and Gabor. **Caltech101-7** contained 1474 images from 7 classes. We adopted six feature types: HOG, GIST, LBP, Gabor, CENTRIST, and Wavelet moments. **3Sources** had six categories. A total of 169 articles were reported by the BBC, Guardian, and Reuters. The term frequency-inverse document frequency (TF-IDF) was utilized to learn 3560-dimensional, 3068-dimensional, and 3631-dimensional features. **BBCSport** included 544 documents, and 3183-dimensional, and 3203-dimensional features were used as the two views.

Table 2
Comparison of clustering performances (mean standard deviation) on five datasets.

DataSet	Metrics	CCA	DCCA	DiMSC	CoMSC	COMVSC	LT-MSC	CVE-AE	MSRLSTR+
ORL	ACC	0.5696(0.005)	0.5964(0.006)	0.8375(0.010)	0.7675(0.002)	0.7925(0.008)	0.7955(0.004)	0.7150(0.008)	0.8381(0.003)
	NMI	0.7601(0.003)	0.7782(0.009)	0.9261(0.011)	0.8722(0.006)	0.9059(0.004)	0.9290(0.003)	0.8762(0.007)	0.9300(0.004)
	F	0.4510(0.010)	0.4771(0.002)	0.7689(0.008)	0.6831(0.007)	0.7334(0.005)	0.7680(0.005)	0.6814(0.010)	0.7702(0.007)
	RI	0.9729(0.003)	0.9740(0.001)	0.9070(0.002)	0.8732(0.001)	0.9268(0.001)	0.9094(0.002)	0.9833(0.009)	0.9892(0.006)
COIL20	ACC	0.5864(0.005)	0.6371(0.008)	0.7081(0.012)	0.6791(0.007)	0.6576(0.008)	0.7342(0.006)	0.7504(0.003)	0.7561(0.002)
	NMI	0.7060(0.004)	0.7599(0.005)	0.8095(0.009)	0.7599(0.009)	0.7959(0.010)	0.8255(0.007)	0.8356(0.006)	0.8362(0.005)
	F	0.5309(0.008)	0.5874(0.011)	0.6543(0.007)	0.6229(0.010)	0.6139(0.011)	0.6938(0.003)	0.7306(0.010)	0.7082(0.006)
	RI	0.9515(0.002)	0.9557(0.003)	0.8455(0.004)	0.8257(0.005)	0.8917(0.005)	0.9686(0.001)	0.9703(0.009)	0.9714(0.001)
Caltech101-7	ACC	0.4535(0.007)	0.5650(0.009)	0.3894(0.008)	0.6988(0.005)	0.7035(0.007)	0.6052(0.006)	0.6092(0.004)	0.7205(0.005)
	NMI	0.5052(0.008)	0.5764(0.014)	0.2751(0.013)	0.5630(0.003)	0.4986(0.014)	0.5427(0.007)	0.6189(0.015)	0.6410(0.008)
	F	0.5351(0.012)	0.6232(0.007)	0.3540(0.005)	0.6025(0.005)	0.6199(0.010)	0.4113(0.002)	0.6369(0.011)	0.7083(0.006)
	RI	0.7325(0.005)	0.7631(0.005)	0.5816(0.003)	0.7052(0.001)	0.7584(0.003)	0.8023(0.003)	0.7859(0.005)	0.8079(0.001)
3Sources	ACC	0.4101(0.009)	0.5217(0.005)	0.7491(0.004)	0.7160(0.003)	0.7160(0.005)	0.7820(0.001)	0.6083(0.009)	0.8220(0.006)
	NMI	0.3564(0.005)	0.5960(0.006)	0.6842(0.001)	0.6057(0.001)	0.6043(0.001)	0.6991(0.005)	0.5759(0.005)	0.7341(0.004)
	F	0.5406(0.002)	0.5862(0.007)	0.7035(0.001)	0.6671(0.004)	0.6686(0.002)	0.7390(0.006)	0.5932(0.004)	0.7102(0.007)
	RI	0.7518(0.001)	0.7697(0.002)	0.8083(0.002)	0.7989(0.002)	0.7667(0.001)	0.8504(0.004)	0.7860(0.001)	0.8780(0.002)
BBCSport	ACC	0.6642(0.005)	0.6616(0.008)	0.8930(0.008)	0.8830(0.006)	0.8983(0.006)	0.7170(0.008)	0.8235(0.008)	0.9414(0.005)
	NMI	0.6966(0.006)	0.6604(0.006)	0.7893(0.004)	0.7820(0.005)	0.7346(0.007)	0.5562(0.010)	0.8075(0.007)	0.8513(0.007)
	F	0.6206(0.009)	0.5909(0.003)	0.8700(0.001)	0.8637(0.001)	0.8322(0.002)	0.6341(0.005)	0.8127(0.002)	0.9202(0.002)
	RI	0.9187(0.004)	0.9136(0.001)	0.9207(0.003)	0.9186(0.002)	0.8703(0.002)	0.8361(0.001)	0.9387(0.001)	0.9624(0.002)

To validate the efficacy of the proposed multi-view subspace representation algorithm, we applied the output of MSRLSTR(Z^v) on a clustering task. The spectral clustering algorithm [46] was applied to similarity matrix S , where $S = \frac{1}{V} \sum_{v=1}^V (|Z^v| + |Z^v|^T) / 2$. We termed the complete pace as MSRLSTR+.

6.1. Clustering performance

We chose advanced multi-view representation and clustering methods to validate the performance of the proposed algorithm. The adopted baselines include CCA [47], DCCA [12], multi-view subspace clustering methods including LT-MSC [34], DiMSC [19], CoMSC [21], and COMVSC [22], and multi-view representation method CVE-AE [15].

In the experiments, we manipulated the accuracy (ACC), normalized mutual information (NMI), Rand index (RI), and F-score (F), as metrics to evaluate the clustering performance. All the metrics ranged from 0 to 1. A better performance is represented by a greater value. These metrics are discussed in the following subsections.

Table 2 lists the clustering performances of the proposed and compared methods. MSRLSTR+ obtained the best performance, except for the F of COIL20 and 3Sources. The proposed model combined the diverse information of a single view and the consistent high-order subspace transformation relationship information of subspace representations, resulting in the complementary and correlation information of multi-view data being fully extracted. Therefore, the proposed method was beneficial for obtaining compelling representations.

MSRLSTR+ yielded better performance than the other subspace clustering methods (DiMSC, CoMSC, and COMVSC) because it achieved the representation based on the subspace transformation relationship rather than the relationship between data objects. The subspace transformation relationship contained high-order information among the subspace representations; this was better for extracting fundamental structural information of multi-view data.

Although these methods could obtain high-level information, MSRLSTR+ exhibited better performance than LT-MSC and CVE-AE. LT-MSC and CVE-AE focused on obtaining high-order correlations between the data objects. However, MSRLSTR+ could accomplish the transformation relationships among subspace representations; this was beneficial for preserving the underlying

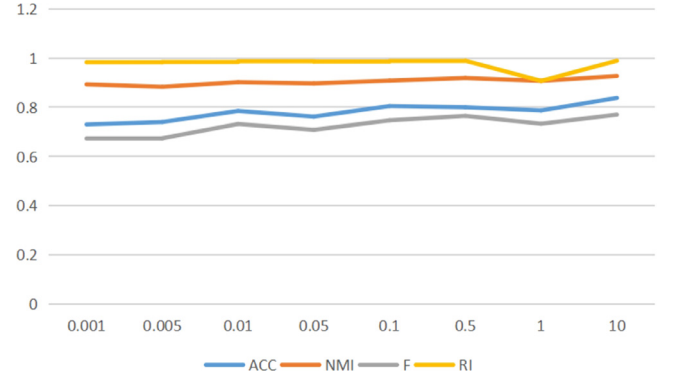


Fig. 4. Effect of different values of λ_1 for clustering performance on the ORL.

structure of the multi-view data. Therefore, high-order relationships among the views were preserved and guaranteed in the final representations.

The performance of MSRLSTR+ was better than that of CCA-based methods, including CCA and DCCA. CCA-based methods focused on learning the consistency representation and neglected the diverse representation between different views. However, MSRLSTR+ extracted both consistent and diverse representations.

6.2. Ablation study

The proposed MSRLSTR contained three main components: manifold, subspace transformation, and low-rank (consistent) constraints. To further verify the effectiveness of each component of the proposed algorithm, we designed an ablation study by tuning regularization parameters λ_1 , λ_2 , and λ_3 . MMR ($\lambda_1 \neq 0$, $\lambda_2 = 0$, and $\lambda_3 = 0$) denotes manifold preserved multi-view representation without a subspace transformation relationship constraint term ($\lambda_2 = 0$) and the consistent constraint term ($\lambda_3 = 0$) in Eq. (5). RMR ($\lambda_1 = 0$, $\lambda_2 \neq 0$, and $\lambda_3 \neq 0$) represents a consistent subspace transformation relationship preserved multi-view representation in Eq. (5) without the manifold constraint. The ablation study was conducted using the ORL, 3Sources, and COIL20 datasets.

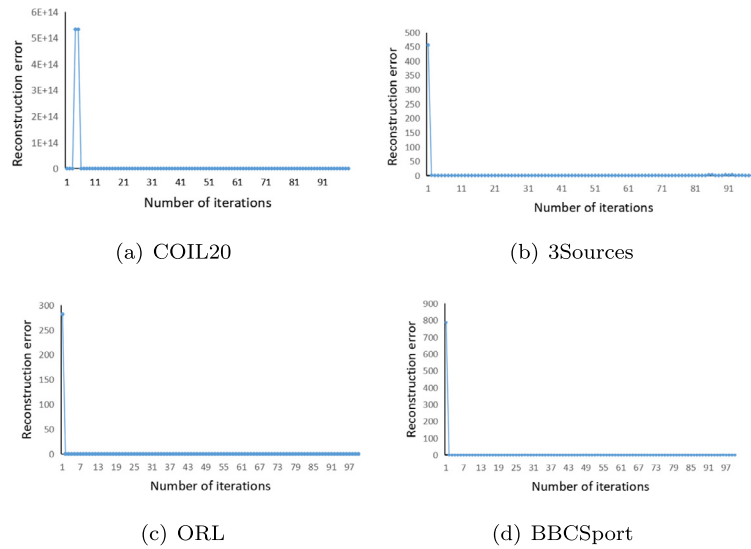


Fig. 5. Performance of convergence of MSRLSTR+ on the COIL20, 3Sources, ORL, and BBCSport.

The MMR has a manifold constraint that can capture the special information of a single view and is helpful for diversity. RMR can capture consistent subspace transformation relationships without requiring diverse information. However, the proposed MSRLSTR+ can capture both diverse information and a consistent subspace transformation relationship. As shown in Fig. 2, MSRLSTR+ outperformed MMR and RMR, indicating that each component of the proposed algorithm was necessary and effective.

6.3. Parameter sensitivity

To verify the impacts of parameters (λ_1 , λ_2 , and λ_3) on the clustering performance, we recorded the clustering metrics (ACC, NMI, F, and RI) with varying λ_1 , λ_2 , and λ_3 on the ORL. In practice, we tuned λ_1 , λ_2 , and λ_3 within the range 0.001, 0.005, 0.01, 0.05, 0.1, 0.5, 1, 10. We first fixed $\lambda_1 = 10$ from experience and analyzed the impact of λ_2 and λ_3 . We then fixed $\lambda_2 = 0.001$ and $\lambda_3 = 1$ and recorded the influence of λ_1 on the metrics.

As shown in Fig. 3, when $\lambda_1 = 10$ was constant, the value of λ_3 varying in the range of 0.5–10 could achieve better performance. The influence of λ_2 was more stable than that of λ_3 . When the $\lambda_2 = 0.001$ and $\lambda_3 = 1$, the proposed algorithm could achieve the best performance. In addition, as observed from Fig. 4, when λ_2 and λ_3 were constant, the clustering metrics were typically stable with the change in λ_1 . Therefore, the clustering performance was insensitive to the value of λ_1 .

6.4. Convergence analysis

The objective function (6) was optimized by the ALM, and it had convexity and could maintain the convergence of MSRLSTR+ to a certain extent. To further verify the convergence of MSRLSTR+, we recorded the reconstruction error of Eq. (6) with iteration on COIL20, 3Sources, ORL, and BBCSport. As shown in Fig. 5, the reconstruction error decreased quickly within a few iterations for all datasets. The convergence performance demonstrated the convergence of the proposed algorithm and proved the effectiveness of the proposed optimization algorithm.

7. Conclusion

This study proposed a novel multi-view unsupervised subspace representation algorithm called MSRLSTR. The relationship between the different views was preserved using a subspace transformation relationship constraint. In addition, a rank constraint was used to ensure the consistency of the different views. The better performance of the proposed MSRLSTR was due to the capturing of the correlation subspace transformation relationship information that is critical to benefit multi-view subspace representation learning. The disadvantage of the proposed MSRLSTR was that the transformation relationship was operated pairwise; this was unfavorable for handling large-scale data clustering. In future work, we will examine the significance of incremental views in multi-view representation learning to further explore the high-order subspace relationship of incremental views.

CRediT authorship contribution statement

Xueying Niu: Writing – original draft, Validation, Formal analysis. **Chaowei Zhang:** Methodology. **Yang Ma:** Writing – review & editing. **Lihua Hu:** Methodology. **Jifu Zhang:** Supervision, Project administration, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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